

IP Issues in CyberSpace: Interface with Patent Law

Ericsson v. Micromax — Delhi High Court, 2013

Ericsson sued Micromax for selling phones that used patented 2G, 3G & EDGE technology, embedded in chips made by Chinese manufacturers.

Micromax didn't write the code. Didn't design the chip. Just sold the product.

The Delhi High Court granted an interim injunction. Micromax couldn't sell those phones until the case was resolved.

They eventually paid royalties.

What is a Patent?

A government-granted monopoly, no one else can make, use, sell, or import your invention for 20 years.

NOVEL	INVENTIVE STEP	INDUSTRIAL USE
No one has done it before <i>Example: A fuel-efficient car engine that no one has built or described before qualifies, but one identical to a prior patent does not.</i>	Not obvious to a skilled professional <i>Example: Combining known engine parts for unexpectedly high efficiency is non-obvious; simply swapping a standard filter is obvious.</i>	Must be practically useful <i>Example: A functional hair-growth serum that grows hair meets utility; a perpetual motion machine claiming impossible energy does not.</i>

In exchange for this monopoly, you must fully disclose how your invention works. After 20 years, it enters the public domain.

Patents, Trademarks and Copyrights



PATENT

The Invention

Protects how something works — a new process, product, or machine.

Protects **Novel inventions & processes**

Registration **Required**

Duration **20 years from filing**

Indian law **Patents Act 1970**

Regulator **Indian Patent Office (IPO)**

Auto-granted? **No — must apply & prove novelty**

Example **A new drug formula, a compression algorithm**



TRADEMARK

The Identity

Protects signs, logos, names, and slogans that identify a brand.

Protects **Brand identity & goodwill**

Registration **Recommended**

Duration **10 years, renewable forever**

Indian law **Trade Marks Act 1999**

Regulator **Trade Marks Registry**

Auto-granted? **Common law rights exist; registration strengthens them**

Example **Amul logo, Tata name, Infosys wordmark**



COPYRIGHT

The Expression

Protects original creative works — writing, art, music, code, films.

Protects **Original creative expression**

Registration **Automatic**

Duration **60 years after author's death**

Indian law **Copyright Act 1957**

Regulator **Copyright Office (under DPIIT)**

Auto-granted? **Yes — at moment of creation**

Example **A novel, a film, a codebase, a song**

3 Ways Engineers Get Into Trouble

01

Building on a Patented Method

You write a compression algorithm. It happens to be patented. You derived it independently — doesn't matter. If you use it commercially, you've infringed.

02

Implementing a Standard (SEPs)

Wi-Fi, Bluetooth, H.264, MP3 — technical standards embed patents. Implementing them without a license = infringement. This is exactly what caught Micromax.

03

Software & Business Method Patents

India bars 'computer programmes per se' under Sec 3(k). But if the software is tied to a technical process or hardware system, it may still be patentable — and US patents are much broader.

IP challenges in digital space

Software patents: the grey zone

01

India's Patents Act §3(k) says software "per se" is not patentable. But what does "per se" mean? Courts have increasingly allowed *computer-implemented inventions (CIIs)* — software that has a technical effect beyond just running on a machine. This mirrors the EU's approach but still lags the US significantly.

Domain names & cybersquatting

02

Registering a domain name identical or confusingly similar to a trademark is "cybersquatting." India addresses this under the IT Act 2000 and the Trade Marks Act 1999. The INDRP (India's domain dispute policy, modeled on ICANN's UDRP) handles .in domain conflicts.

AI & machine learning patents

03

Can an AI be an inventor? India currently says no — only a natural person can be listed. This creates a gap as companies file AI-generated innovations under human names, raising authenticity questions.

IP challenges in digital space

Open source vs patent pools

04

When a company uses open-source software that unknowingly infringes a patent, liability is complex. Patent pools (like those for Wi-Fi, HEVC, MP3) attempt to create licensing ecosystems — but India's SMEs are often blind to these until litigation hits.

Data as intellectual property

05

Raw data cannot be patented. But compiled datasets with significant investment ("database rights" in the EU) have quasi-IP protection. India lacks a specific database protection law — companies rely on trade secrets and contracts instead.

The Patents Act, 1970 — Key Sections

Sec 2(1)(j)

Definition of Invention

New product or process with an inventive step and industrial application.

Sec 84

Compulsory Licensing

If patent holder doesn't make the invention available at reasonable price, anyone can apply for a compulsory license.

Sec 3(k)

Not Patentable

Algorithms, mathematical methods, business methods, computer programmes per se — excluded.

Sec 104–114

Infringement & Remedies

Filed in District Courts or High Courts. Remedies: injunctions, damages, accounts of profits.

International Frameworks

TRIPS Agreement

Trade-Related Aspects of Intellectual Property Rights

India is a WTO signatory. TRIPS sets minimum IP standards globally — patents must be available in all tech fields for at least 20 years.

India amended its Patents Act in 2005 to comply with TRIPS — which is when tech and pharma patents became properly enforceable here.

Patent Cooperation Treaty (PCT)

Administered by WIPO

One international application = protection request in multiple countries simultaneously.

Why this matters: A US patent filed under PCT can have an Indian equivalent. When you use an open-source library from the US, the underlying patents may already apply in India.

The Software Patent Problem

The line between a patentable invention and a non-patentable algorithm is drawn by how a lawyer writes the claim, not by what the invention actually is.

Case Example: Amazon One-Click Patent (US, 1999)

The 'invention': If a user has previously stored payment and address info, clicking a single button completes a purchase without reconfirmation.

Amazon sued Barnes & Noble, who had implemented something similar. Barnes & Noble had to change their checkout flow mid-litigation.

Is a checkout shortcut worth 20 years of monopoly? Reasonable people disagree. But it was enforceable.

A patent on a training process or architecture, not just the model, can still be infringed if you deploy it commercially.

Patent Trolls & Structural Problems

What is a Patent Troll?

A company that exists solely to hold patents and sue infringers. They produce nothing. They build nothing.

They buy patents cheaply from bankrupt companies or inventors, then send demand letters to businesses.

Even a baseless suit costs money to defend. Companies often settle just because fighting is more expensive.

The Real Tension

Patents are meant to reward disclosure and fund innovation. After 20 years, the invention becomes public.

In tech, 20 years is an eternity. By the time it expires, the tech is obsolete.

In practice: patents create barriers to entry for smaller players and give incumbents tools to slow competitors.

What You Should Actually Do as an Engineer

1

Before using any library or third-party component, check its license for patent grants. Apache 2.0 includes an explicit patent grant. MIT does not.

2

If you build something novel, document everything with dates. Records of invention matter for establishing prior art or claiming inventorship.

3

Read your employment contract. Any invention you create in the course of employment typically belongs to the employer — Sec 6 & 7, Patents Act.

4

Do not assume complexity = protection. And do not assume obviousness = no patent. Both assumptions have burned companies.

Further Reading

Ericsson v. Micromax

Delhi High Court, 2013 — India

SEPs · FRAND Licensing · Injunction

The hook case. Standard Essential Patents, what happens when you use tech in a standard without a license.

Novartis AG v. Union of India

Supreme Court of India, 2013

Sec 3(d) · Evergreening · TRIPS Compliance

Pharmaceutical, but sets the interpretive framework for what counts as a genuine 'inventive step' under Indian law.

Amazon v. Barnesandnoble.com

US Court of Appeals, 2001

Software Patents · One-Click · Enforceability

The one-click patent. Software patent enforceability and how absurd the system can get in practice.

Bilski v. Kappos

US Supreme Court, 2010

Business Method Patents · Software

Defines limits on business method patents. Useful for understanding the global software patent debate.